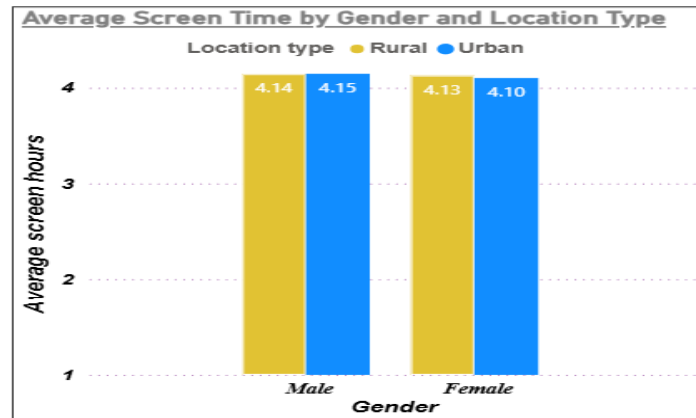


Screen Sense: Kids' Screentime Visualisation

Week 5 and Week 6:

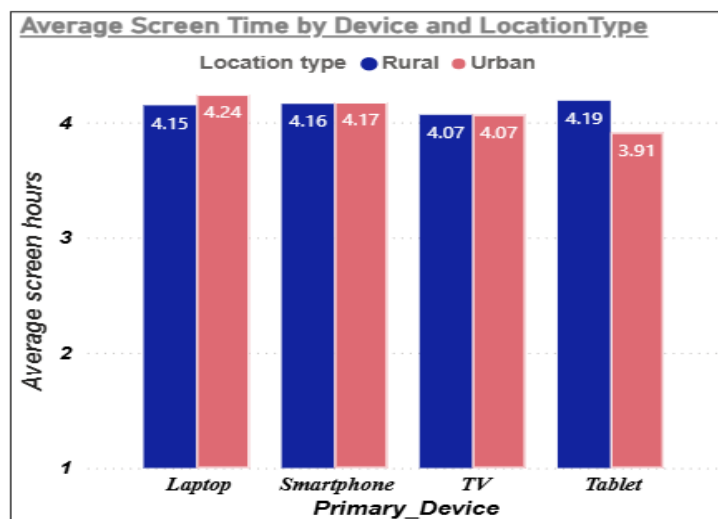
Cohort and Segment Analysis:

1. Average Screen Time by Gender and Location Type



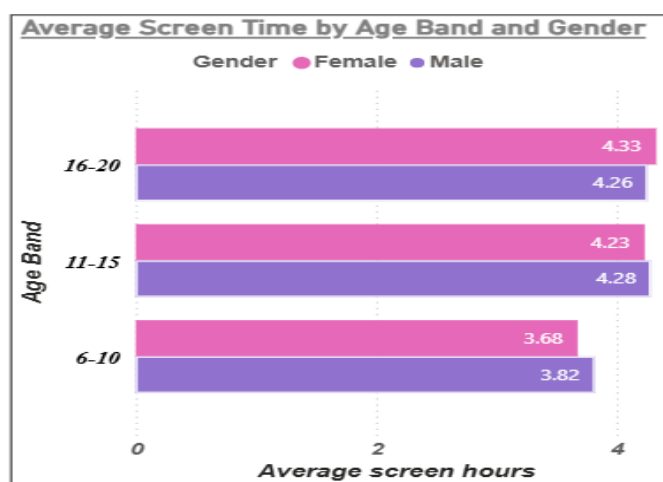
- This bar chart compares the average screen time between rural and urban users across genders. Both male and female users show nearly similar screen usage, with only slight differences between locations. Urban males record the highest usage at 4.15 hours, while urban females show slightly lower usage at 4.10 hours. The visualisation highlights that location type has minimal impact on screen time. Overall, gender-based screen habits remain largely regardless of rural or urban background.

2. Average Screen Time by Device and Location Type



- This chart compares average screen time across different devices for rural and urban users. Among all devices, **urban laptop usage is the highest** (≈ 4.24 hours), indicating strong dependence on laptops in urban areas. The **lowest usage overall is urban tablet usage** (≈ 3.91 hours), showing reduced interest in tablets among urban users. Smartphone and TV usage remain almost the same in both locations, reflecting consistent habits across regions. Tablet usage shows the biggest location gap, with **rural users spending more time** (≈ 4.19 hours) compared to urban users. Overall, the chart highlights that device preferences are largely similar across locations, with minor variations—urban users lean slightly more toward laptops, while rural users use tablets more.

3. Average Screen Time by Age Band and Gender



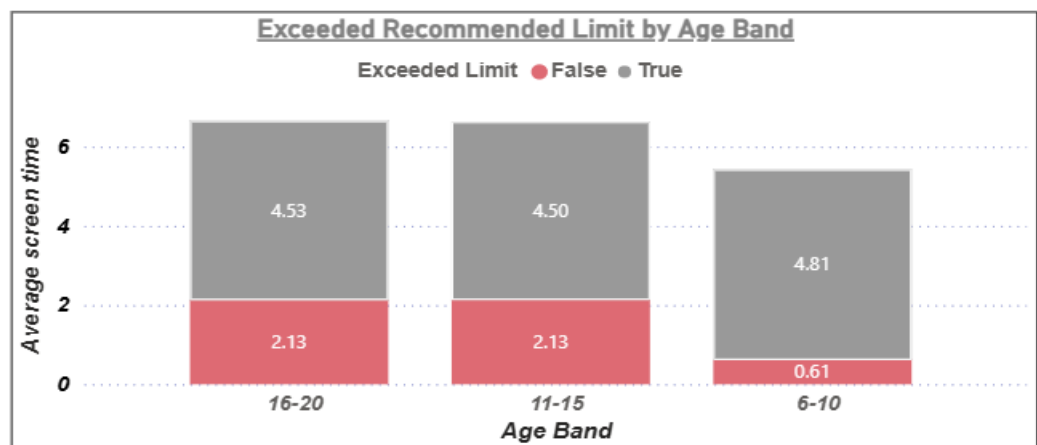
- This chart shows how screen time varies across age groups for male and female users. The **highest screen time overall** comes from **females aged 16–20** (4.33 hrs). Males in the same age group follow close behind at 4.26 hours. The **lowest screen time** is for **females aged 6–10** (3.68 hrs), with males slightly higher at 3.82 hours. Screen time clearly increases with age, peaking in the teen years. Age is the strongest factor influencing screen time, with older students consistently spending more time on screens than younger ones.

4. Average Screen Time by Age Band and Device (Heatmap)

Average Screen time by Age Band and Device				
Age_Band	Laptop	Smartphone	Tablet	TV
11-15	4.15	4.29	4.22	4.31
16-20	4.33	4.28	4.26	4.30
6-10		3.78	3.72	3.74

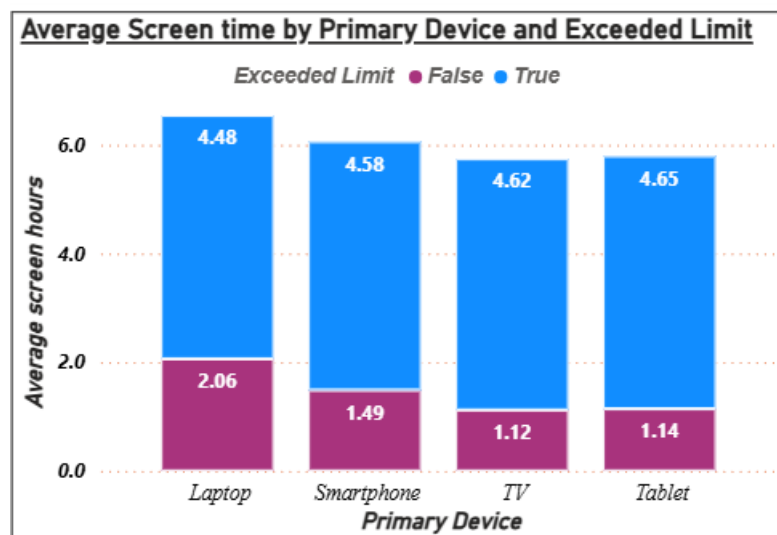
- This heatmap compares device usage across different age groups, with darker colours showing higher screen time. The **highest usage overall** appears in the **16–20 age group**, especially on laptops (4.33 hrs) and smartphones (4.28 hrs). The **11–15 group** also shows high engagement, particularly with smartphones (4.29 hrs) and TVs (4.31 hrs). The **lowest overall usage** comes from the **6–10 age group**, especially tablet usage (3.72 hrs). Device preferences remain consistent across age groups, but older users consistently spend more time on every device. Screen time increases with age regardless of device type, and teens (16–20) are the most active digital users across all platforms.

5. Exceeded Recommended Limit by Age Band



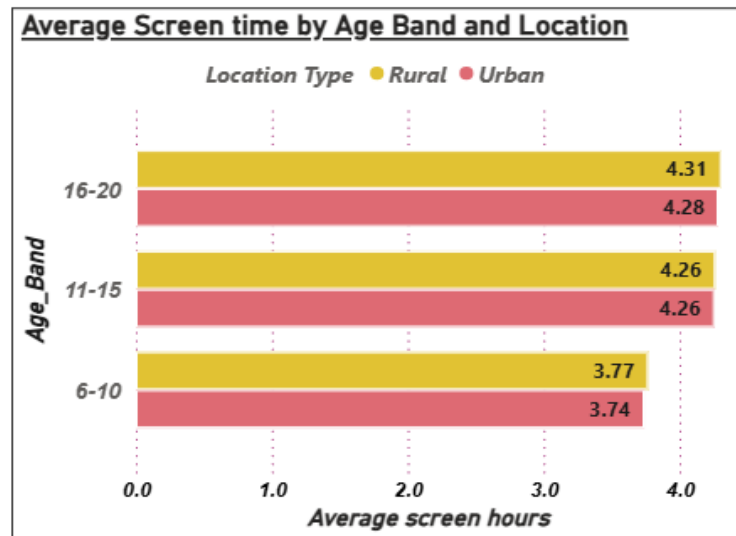
- This stacked chart shows how much screen time each age group spends, separating **safe usage (False)** from **unsafe overuse (True)**. The **16–20** and **11–15** age groups show the **highest unsafe screen time**, each spending **2.13 hours more than the recommended healthy limit**. This indicates a clear pattern of **excessive and unhealthy screen exposure among teens**. The **6–10** age group shows much **lower excess usage**, with only **0.61 hours** of unsafe screen time, but still exceeds the recommended limit. Overall, the chart highlights that **older children and teenagers are the most at risk**, spending significantly more time beyond safe screen-time guidelines.

6. Average Screen Time by Primary Device and Exceeded Limit



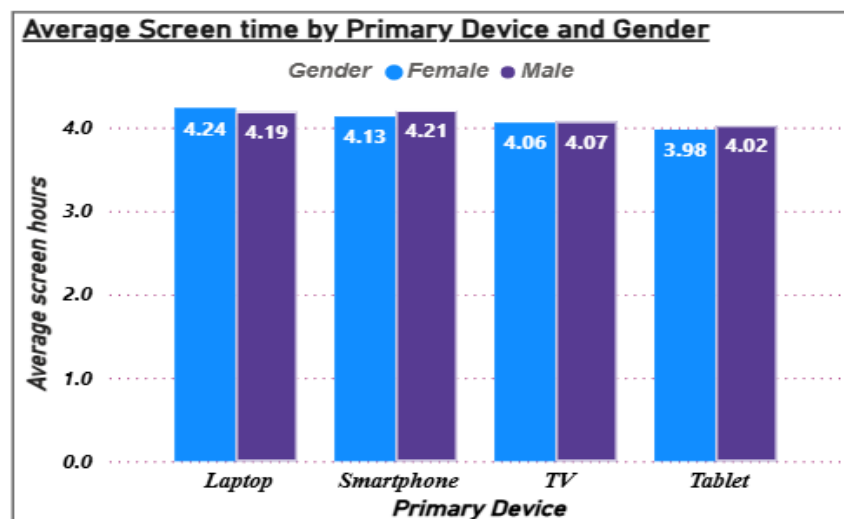
- This chart shows how often users exceed the recommended screen-time limit for each device type. A value of **TRUE** means the user has gone **beyond healthy screen-time levels**, indicating overuse. Smartphone users exceed the limit most frequently, followed by laptop and tablet users, while TV show the lowest exceed rate. The pattern suggests that portable and highly engaging devices like smartphones lead to higher chances of unhealthy usage. Overall, this helps identify which devices contribute most to excessive screen time and where digital-wellbeing interventions may be needed.

7. Average Screen Time by Age Band and Location



- This chart compares screen time across age groups in rural and urban areas. The 16–20 age band has the highest usage, with rural (4.31 hrs) slightly higher than urban (4.28 hrs). Younger children (6–10) show the lowest values around 3.74–3.77 hrs. Differences between locations are small but consistent across all age groups. Overall, age is a stronger driver of screen time than location.

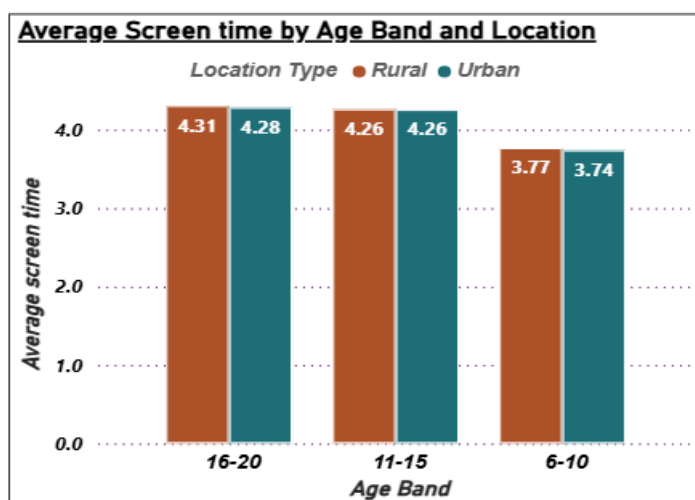
8. Average Screen Time by Primary Device and Gender



- This bar chart compares device usage between males and females. Laptop usage is highest for both genders, with females at 4.24 hrs

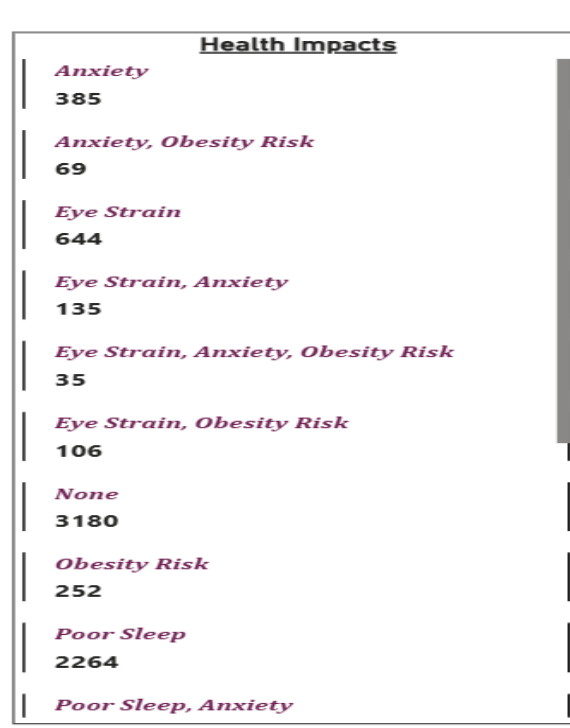
and males at 4.19 hrs. Females show slightly higher screen time across most devices except tablets. The smallest difference is seen in smartphone and TV usage. Overall, gender differences are small, but females show marginally higher screen time.

9. Average Screen Time by Age Band and Location



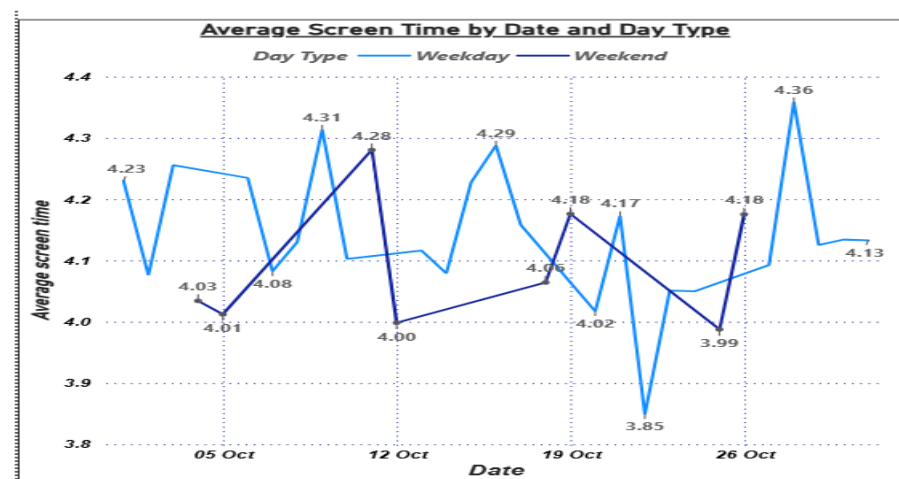
- Teens aged **16–20** show the **highest screen time**, with rural (4.31 hrs) slightly higher than urban (4.28 hrs). The **11–15 age group** also maintains high screen use, with rural and urban almost identical (4.26 hrs each). Younger children **6–10** have noticeably lower screen time, around **3.7 hours** for both locations. Screen time consistently **increases with age**, peaking in late teens regardless of location.

10. Health Impacts (List Visual)



- This list visual displays the distribution of different health issues related to screen time. Most users report **no health issues** (3180), followed by high counts of **eye strain** (644) and **poor sleep** (2264). Combinations like “Eye Strain & Anxiety” and “Obesity Risk” appear in smaller numbers. Severe combined impacts like “Eye Strain, Anxiety, Obesity Risk” are much less common. Overall, eye strain and sleep problems are the most significant screen-related concerns.

11. Average Screen Time by Date and Day Type (Line Chart)



- The chart compares **weekday vs. weekend** screen time across October. **Weekend screen time is generally higher**, with peaks like **4.31 hrs (12 Oct)** and **4.36 hrs (26 Oct)**. Weekday screen time stays slightly **lower and more stable**, mostly between **4.00–4.18 hrs**. A noticeable dip occurs on **21 Oct** (3.85 hrs), which is the **lowest screen time** in the month. Overall, users spend **more time on screens during weekends**, showing relaxed day usage patterns.

12. Screen Time by Health Impacts and Device Types (Heatmap Table)

Screen Time by Health Impacts and Device Types					
Health_Impacts	Laptop	Smartphone	Tablet	TV	Total
Anxiety	62	180	40	103	385
Anxiety, Obesity Risk	11	34	8	16	69
Eye Strain	101	307	66	170	644
Eye Strain, Anxiety	28	63	19	25	135
Eye Strain, Anxiety, Obesity Risk	7	19		9	35
Eye Strain, Obesity Risk	16	57	10	23	106
None	379	1445	459	897	3180
Obesity Risk	37	132	26	57	252
Poor Sleep	370	1049	271	574	2264
Poor Sleep, Anxiety	107	272	77	151	607
Poor Sleep, Anxiety, Obesity Risk	15	35	11	17	78
Poor Sleep, Eye Strain	170	489	110	209	978
Poor Sleep, Eye Strain, Anxiety	39	123	34	62	258
Poor Sleep, Eye Strain, Anxiety, Obesity Risk	10	17	4	6	37
Poor Sleep, Eye Strain, Obesity Risk	25	97	24	42	188
Poor Sleep, Obesity Risk	54	221	60	117	452
Total	1431	4540	1219	2478	9668

- The table shows how different **health issues correlate with screen time** across devices. The largest group is **“None” (3180 users)**, with the highest smartphone use (1445), meaning most users report **no major health impact**. **Eye strain (644 users)** is the most common issue among affected users, especially linked to **smartphone (307)** and **TV (170)** usage. **Poor Sleep (2264 users)** also shows high counts, particularly with smartphones (1049) and TVs (574). The heatmap reveals **smartphones contribute most** to health-related screen exposure, followed by **TVs**, while **tablets** contribute the least.

